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Light-receiving device and method for manufacturing light-receiving device

Abstract

A light-receiving device includes a light-receiving element in which a plurality of light-receiving portions are provided; and a glass cover that has a first surface and a second surface opposite to the first surface and that is fixed above the light-receiving element such that the second surface faces the light-receiving element. The glass cover includes a plurality of diffraction gratings provided in at least one of the first surface and an inside of the glass cover. Each of the plurality of diffraction gratings is configured to concentrate laser light for distance measurement on each of the plurality of light-receiving portions, the laser light for distance measurement incident on the first surface of the glass cover.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a light-receiving device and a method for manufacturing a light-receiving device.

BACKGROUND ART

(2) As a technique for measuring the distance to an object using a time of flight (TOF) method, a light detection and ranging (LiDAR) system has attracted attention. For this reason, various technical developments have been made on a light-receiving device used in the LiDAR system (for example, refer to Patent Literature 1).

CITATION LIST

Patent Literature

(3) Patent Literature 1: Japanese Unexamined Patent Publication No. 2017-224755

SUMMARY OF INVENTION

Technical Problem

(4) In the LiDAR system, it is important how accurately the light-receiving device can receive laser light for distance measurement that is emitted from a light source and is reflected by an object. Meanwhile, it is not desirable that the thickness of the light-receiving device is increased to accurately receive the laser light for distance measurement.

(5) An object of the present disclosure is to provide a light-receiving device that can accurately receive laser light for distance measurement and can be made thin, and a method for manufacturing such a light-receiving device.

Solution to Problem

(6) A light-receiving device according to one aspect of the present disclosure includes: a light-receiving element in which a plurality of light-receiving portions are provided; and a glass cover that has a first surface and a second surface opposite to the first surface and that is fixed above the light-receiving element such that the second surface faces the light-receiving element. The glass cover includes a plurality of diffraction gratings provided in at least one of the first surface and an inside of the glass cover. Each of the plurality of diffraction gratings is configured to concentrate laser light for distance measurement on each of the plurality of light-receiving portions, the laser light for distance measurement incident on the first surface of the glass cover.

(7) In the light-receiving device, the glass cover includes the plurality of diffraction gratings, and each of the plurality of diffraction gratings is configured to concentrate the laser light for distance

measurement on each of the plurality of light-receiving portions, the laser light for distance measurement incident on the glass cover. Accordingly, a reduction in crosstalk between the adjacent light-receiving portions and an improvement in sensitivity of each of the plurality of light-receiving portions can be achieved. Moreover, the plurality of diffraction gratings are provided in at least one of the first surface and the inside of the glass cover. Accordingly, it is not necessary to provide, as a separate member, a light-concentrating element between the light-receiving element and the glass cover. In addition, since the distance from the light-receiving element to the plurality of diffraction gratings is larger than that when the plurality of diffraction gratings are provided in the second surface of the glass cover, the glass cover can be fixed above the light-receiving element in a state where the glass cover is placed closer to the light-receiving element. As a result, the light-receiving device can accurately receive the laser light for distance measurement, and can be made thin.

(8) In the light-receiving device according to one aspect of the present disclosure, the glass cover may be fixed above the light-receiving element in a state where the glass cover is separated from the light-receiving element. Accordingly, the distance from the light-receiving element to the plurality of diffraction gratings can be set to an appropriate distance, and the laser light for distance measurement incident on the glass cover can be reliably concentrated on each of the plurality of light-receiving portions.

(9) In the light-receiving device according to one aspect of the present disclosure, the glass cover may be fixed on the light-receiving element in a state where the glass cover is in contact with the light-receiving element. Accordingly, the device can be made thinner.

(10) The light-receiving device according to one aspect of the present disclosure may further include a light source that emits the laser light for distance measurement. Each of the plurality of diffraction gratings may be configured to concentrate the laser light for distance measurement on each of the plurality of light-receiving portions, the laser light for distance measurement emitted from the light source and reflected by an object. Accordingly, the distance to the object can be accurately measured.

(11) In the light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate and a first film provided on a surface on a first surface side of the glass substrate. The plurality of diffraction gratings may be formed in the first film. Accordingly, the distance from the light-receiving element to the plurality of diffraction gratings can be set to an appropriate distance, and the laser light for distance measurement incident on the glass cover can be reliably concentrated on each of the plurality of light-receiving portions.

(12) In the light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate, and the plurality of diffraction gratings may be formed in a surface on a first surface side of the glass substrate. Accordingly, the distance from the light-receiving element to the plurality of diffraction gratings can be set to an appropriate distance, and the laser light for distance measurement incident on the glass cover can be reliably concentrated on each of the plurality of light-receiving portions.

(13) In the light-receiving device according to one aspect of the present disclosure, the glass cover may further include a filter disposed on a side opposite to the light-receiving element with respect to the plurality of diffraction gratings. The filter may have transmission properties for the laser light for distance measurement, and have light shielding properties for light having wavelengths other than a wavelength of the laser light for distance measurement. In such a manner, since the filter is provided in a previous stage of the plurality of diffraction gratings, the light having the wavelengths other than the wavelength of the laser light for distance measurement can be reliably cut, and the SN ratio of the light-receiving element can be improved.

(14) The light-receiving device according to one aspect of the present disclosure may further include a substrate that has an inside surface on which the light-receiving element is disposed, an outside surface opposite to the inside surface, and a through-hole being open to the inside surface

and the outside surface and that forms, together with the glass cover, a package that houses the light-receiving element; a metallic film provided on an inner surface of the through-hole; a metallic ball disposed in the through-hole; and a solder member that is disposed between the metallic film and the metallic ball in the through-hole and that fixes the metallic ball in the through-hole. The through hole may include an inside opening portion located in the inside surface, an outside opening portion located in the outside surface, and an intermediate opening portion located between the inside surface and the outside surface. The outside opening portion may have a shape through which the metallic ball is passable. The intermediate opening portion may have a shape through which the metallic ball is not passable. The metallic film may include a cylindrical portion provided on a side surface of the inner surface of the through-hole, the side surface being located between the outside opening portion and the intermediate opening portion. The metallic ball may be disposed in the through-hole to be surrounded by the cylindrical portion. The solder member may seal an area between the cylindrical portion and the metallic ball in a state where the solder member extends from the metallic ball to the cylindrical portion. Accordingly, during manufacturing of the light-receiving device, for example, while degassing the inside of the package through the through-hole, the metallic ball can be fixed in the through-hole by the solder member, and the area between the cylindrical portion of the metallic film and the metallic ball can be sealed by the solder member. At this time, since at least a part of the intermediate opening portion of the through-hole is closed by the metallic ball, the molten solder member can be suppressed from infiltrating into the package. Further, the intermediate opening portion of the through-hole has a shape through which the metallic ball is not passable, and in a state where the solder member extends from the metallic ball to the cylindrical portion of the metallic film, the area between the cylindrical portion of the metallic film and the metallic ball is sealed by the solder member. Accordingly, in the manufactured light-receiving device, the fixed state of the metallic ball is stabilized, so that the sealing state of the through-hole can be reliably maintained.

(15) In the light-receiving device according to one aspect of the present disclosure, the intermediate opening portion may have a long shape. Accordingly, during manufacturing of the light-receiving device, when the metallic ball is fixed in the through-hole by the solder member, a part of the intermediate opening portion of the through-hole is not closed by the metallic ball, so that the inside of the package can be reliably degassed through the through-hole.

(16) In the light-receiving device according to one aspect of the present disclosure, the through-hole may be a tapered hole that widens from the inside surface toward the outside surface. Accordingly, the through-hole including the outside opening portion having a shape through which the metallic ball is passable and the intermediate opening portion having a shape through which the metallic ball is not passable can be easily and reliably obtained.

(17) In the light-receiving device according to one aspect of the present disclosure, the through-hole may be a stepped hole that is widened in width on an outside surface side. Accordingly, the through-hole including the outside opening portion having a shape through which the metallic ball is passable and the intermediate opening portion having a shape through which the metallic ball is not passable can be easily and reliably obtained.

(18) In the light-receiving device according to one aspect of the present disclosure, the metallic film may further include a flange portion provided on the outside surface to extend along an outer edge of the outside opening portion. The cylindrical portion may be connected to the flange portion. Accordingly, when a mounting substrate on which the light-receiving device is to be mounted is provided with a metallic pattern corresponding to the outside opening portion of the through-hole and the flange portion of the metallic film, and the flange portion is joined to the metallic pattern, even if the metallic ball falls on the metallic pattern during joining, the sealing state of the through-hole in the light-receiving device after mounting can be reliably maintained.

(19) In the light-receiving device according to one aspect of the present disclosure, a surface on an outside opening portion side of the solder member may be a curved surface that is recessed to an

inside opening portion side in the area between the cylindrical portion and the metallic ball. Accordingly, the local concentration of stress can be suppressed from occurring in the solder member, and the sealing state of the through-hole can be more reliably maintained.

(20) According to one aspect of the present disclosure, there is provided a method for manufacturing a light-receiving device, the method including: a first step of preparing a light-receiving element in which a plurality of light-receiving portions are provided, and a glass cover having a first surface and a second surface opposite to the first surface; a second step of disposing the glass cover above the light-receiving element such that the second surface faces the light-receiving element, after the first step; and a third step of forming a plurality of light-concentrating elements in the glass cover by irradiating the glass cover with laser light for processing after the second step. In the third step, each of the plurality of light-concentrating elements is formed such that the plurality of light-concentrating elements concentrate laser light for distance measurement on each of the plurality of light-receiving portions, the laser light for distance measurement incident on the first surface of the glass cover.

(21) In the method for manufacturing the light-receiving device, the plurality of light-concentrating elements are formed in the glass cover by disposing the glass cover above the light-receiving element and then irradiating the glass cover with the laser light for processing. Accordingly, the light-receiving device in which each of the plurality of light-concentrating elements is accurately positioned with respect to each of the plurality of light-receiving portions can be efficiently obtained. As a result, according to the method for manufacturing the light-receiving device, the light-receiving device described above can be appropriately manufactured.

(22) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the second step, the glass cover may be fixed above the light-receiving element such that the glass cover is separated from the light-receiving element. Accordingly, in the manufactured light-receiving device, the distance from the light-receiving element to the plurality of light-concentrating elements can be set to an appropriate distance, and the laser light for distance measurement incident on the glass cover can be reliably concentrated on each of the plurality of light-receiving portions.

(23) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the second step, the glass cover may be fixed on the light-receiving element such that the glass cover is in contact with the light-receiving element. Accordingly, the manufactured light-receiving device can be made thinner.

(24) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate and a first film provided on a surface on a first surface side of the glass substrate. In the third step, the plurality of light-concentrating elements may be formed in the first film. Accordingly, the influence of the irradiation of the laser light for processing on the light-receiving element can be suppressed.

(25) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the plurality of light-concentrating elements may be a plurality of diffraction gratings. Accordingly, the plurality of light-concentrating elements can be easily provided in the glass cover by the irradiation of the laser light for processing.

(26) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate. In the third step, the plurality of light-concentrating elements may be formed in the glass substrate. Accordingly, various light-concentrating elements can be provided in the glass cover by the irradiation of the laser light for processing.

(27) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the third step, the plurality of light-concentrating elements may be formed in a surface on a first surface side of the glass substrate. Accordingly, the influence of the irradiation of the laser light for processing on the light-receiving element can be suppressed.

(28) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the plurality of light-concentrating elements may be a plurality of diffraction gratings. Accordingly, the plurality of light-concentrating elements can be easily provided in the glass cover by the irradiation of the laser light for processing.

(29) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the third step, the plurality of light-concentrating elements may be formed inside the glass substrate. Accordingly, the influence of the irradiation of the laser light for processing on the light-receiving element can be suppressed.

(30) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the plurality of light-concentrating elements may be a plurality of diffraction gratings, a plurality of lens surfaces, or a plurality of light-guiding portions. Accordingly, appropriate light-concentrating elements according to the application can be provided in the glass cover.

(31) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate and a second film provided on a surface on a second surface side of the glass substrate. The second film may have transmission properties for the laser light for distance measurement, and have light shielding properties for the laser light for processing. Accordingly, the influence of the irradiation of the laser light for processing on the light-receiving element can be more reliably suppressed.

(32) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, the glass cover may include a glass substrate. The glass substrate may have transmission properties for the laser light for distance measurement, and have light shielding properties for the laser light for processing. Accordingly, the influence of the irradiation of the laser light for processing on the light-receiving element can be more reliably suppressed.

(33) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the first step, a semiconductor wafer including a plurality of portions that become the light-receiving elements, and a glass wafer including a plurality of portions that become the glass covers may be prepared. In the second step, the glass wafer may be fixed on the semiconductor wafer. In the third step, the plurality of light-concentrating elements may be formed for each of the portions, which become the light-receiving elements, by irradiating the glass wafer with the laser light for processing. Accordingly, the plurality of light-concentrating elements can be efficiently formed by the irradiation of the laser light for processing.

(34) The method for manufacturing a light-receiving device according to one aspect of the present disclosure may further include a fourth step of cutting the semiconductor wafer and the glass wafer for each of the portions that become, which the light-receiving elements, after third step. Accordingly, a plurality of the light-receiving devices can be efficiently manufactured.

(35) According to one aspect of the present disclosure, there is provided a method for manufacturing a light-receiving device, which is a method for manufacturing the light-receiving device described above, the method including: a fifth step of disposing the light-receiving element on the inside surface of the substrate in which the metallic film is provided on the inner surface of the through-hole, so that a package which houses the light-receiving element is formed by the glass cover and the substrate; and a sixth step of forming the solder member by disposing a solder ball, which includes the metallic ball and a solder layer covering the metallic ball, in the through-hole from the outside opening portion and melting the solder layer after the fifth step.

(36) According to the method for manufacturing the light-receiving device, the through-hole can be reliably sealed while suppressing the infiltration of the molten solder layer into the package.

(37) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the sixth step, the solder layer may be melted by irradiating the solder ball with laser light. Accordingly, the solder layer can be locally heated, so that thermal influence on the light-receiving device and the like can be suppressed.

(38) In the method for manufacturing a light-receiving device according to one aspect of the present disclosure, in the sixth step, the solder layer may be melted while an inside of the package is degassed through the through-hole. Accordingly, the inside of the package can be reliably degassed through the through-hole while suppressing the infiltration of the molten solder layer into the package.

Advantageous Effects of Invention

(39) According to the present disclosure, it is possible to provide the light-receiving device that can accurately receive the laser light for distance measurement and can be made thin, and the method for manufacturing such a light-receiving device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a configuration view of a distance measurement system including a light-receiving device of a first embodiment.
- (2) FIG. 2 is a perspective view of the light-receiving device illustrated in FIG. 1.
- (3) FIG. 3 is an enlarged view of the light-receiving device illustrated in FIG. 2.
- (4) FIG. 4 is a view illustrating one step of a method for manufacturing the light-receiving device illustrated in FIG. 2.
- (5) FIG. 5 is a view illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 2.
- (6) FIG. 6 is a view illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 2.
- (7) FIG. 7 is a perspective view of a light-receiving device of a second embodiment.
- (8) FIG. 8 is a view illustrating one step of a method for manufacturing the light-receiving device illustrated in FIG. 7.
- (9) FIG. 9 is a view illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 7.
- (10) FIG. 10 is a view illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 7.
- (11) FIG. 11 is a view illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 7.
- (12) FIG. 12 is a perspective view of a light-receiving device of a third embodiment.
- (13) FIG. 13 provides cross-sectional views of a portion of a printed wiring substrate of the light-receiving device illustrated in FIG. 12.
- (14) FIG. 14 is a bottom view of a portion of the printed wiring substrate illustrated in FIG. 13.
- (15) FIG. 15 provides views illustrating one step of a method for manufacturing the light-receiving device illustrated in FIG. 12.
- (16) FIG. 16 provides views illustrating one step of the method for manufacturing the light-receiving device illustrated in FIG. 12.
- (17) FIG. 17 provides views illustrating one step of a method for mounting the light-receiving device illustrated in FIG. 12.
- (18) FIG. 18 provides views illustrating one step of the method for mounting the light-receiving device illustrated in FIG. 12.
- (19) FIG. 19 is an enlarged view of a glass cover of a first modification example.
- (20) FIG. 20 is an enlarged view of a glass cover of a second modification example.
- (21) FIG. 21 is an enlarged view of a glass cover of a third modification example.
- (22) FIG. 22 is an enlarged view of a glass cover of a fourth modification example.
- (23) FIG. 23 provides cross-sectional views of a glass cover of a fifth modification example and a

glass cover of a sixth modification example.

(24) FIG. 24 provides cross-sectional views of a portion of a printed wiring substrate of a light-receiving device of a modification example.

(25) FIG. 25 is a bottom view of a portion of the printed wiring substrate illustrated in FIG. 24.

DESCRIPTION OF EMBODIMENTS

(26) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. Incidentally, in the drawings, the same or equivalent portions are denoted by the same reference signs, and duplicated portions will be omitted.

First Embodiment

(27) As illustrated in FIG. 1, a distance measurement system **100** includes a light source **10** and a light-receiving device **1A**. The light source **10** emits laser light **L1** for distance measurement toward an object **A**. The light source **10** is formed of, for example, a light-emitting element such as a light-emitting diode or a semiconductor laser. The light-receiving device **1A** receives the laser light **L1** for distance measurement that is emitted from the light source **10** and is reflected by the object **A**. The distance measurement system **100** is a light detection and ranging (LiDAR) system that measures the distance to the object **A** using a time of flight (TOF) method.

(28) As illustrated in FIG. 2, the light-receiving device **1A** includes a printed wiring substrate **2**, a light-receiving element **3**, a frame **4**, and a glass cover **5**. In the following description, a direction in which the laser light **L1** for distance measurement is incident on the light-receiving device **1A** is referred to as a Z-axis direction, one direction perpendicular to the Z-axis direction is referred to as an X-axis direction, and a direction perpendicular to the Z-axis direction and the X-axis direction is referred to as a Y-axis direction. Incidentally, in the distance measurement system **100**, since the distance from the light-receiving device **1A** to the object **A** is much larger than the size of the light-receiving device **1A**, it can be considered the laser light **L1** for distance measurement is incident on the light-receiving device **1A** in parallel with the Z-axis direction.

(29) The printed wiring substrate **2** is formed in, for example, a rectangular plate shape. As one example, the length of the printed wiring substrate **2** in the X-axis direction is approximately 20 mm, the length of the printed wiring substrate **2** in the Y-axis direction is approximately 10 mm, and the thickness of the printed wiring substrate **2** is approximately 1 mm.

(30) The light-receiving element **3** is mounted on an inside surface **2a** on a light-incident side (side on which the laser light **L1** for distance measurement is incident) of the printed wiring substrate **2**. The light-receiving element **3** is provided with a plurality of light-receiving portions **3a**. In the present embodiment, the light-receiving element **3** is a surface incidence type linear sensor including the plurality of light-receiving portions **3a** that are arranged in one dimension along the X-axis direction. Each of the light-receiving portions **3a** is formed of, for example, an avalanche photodiode, and functions as a pixel. Each of the light-receiving portions **3a** is formed in, for example, a rectangular shape. As one example, the length of each of the light-receiving portions **3a** in the X-axis direction is 0.45 mm, the length of each of the light-receiving portions **3a** in the Y-axis direction is 0.46 mm, and the center-to-center distance (pitch) of the adjacent light-receiving portions **3a** is 0.55 mm. Incidentally, an electronic component **6** such as an application specific integrated circuit (ASIC) is also mounted on the inside surface **2a** of the printed wiring substrate **2**.

(31) The frame **4** is fixed to the inside surface **2a** of the printed wiring substrate **2** in a state where the frame **4** surrounds the light-receiving element **3** and the electronic component **6** when viewed in the Z-axis direction. The frame **4** is fixed to the inside surface **2a** of the printed wiring substrate **2** by, for example, a resin sheet. The frame **4** is a frame-shaped support member extending along an outer edge of the printed wiring substrate **2**. The height of a surface **4a** on the light-incident side of the frame **4** (distance from the inside surface **2a** of the printed wiring substrate **2**) is larger than the height of the light-receiving element **3** and the height of the electronic component **6**. For example, the frame **4** is formed in a rectangular frame shape from a glass epoxy material. As one example, the length of the frame **4** in the X-axis direction is approximately 20 mm, the length of the frame **4**

in the Y-axis direction is approximately 10 mm, and the thickness of the frame **4** is approximately 1 mm.

(32) The glass cover **5** is formed of a glass substrate **50**. The glass substrate **50** is formed in, for example, a rectangular plate shape from a glass material having transmission properties for the laser light **L1** for distance measurement. As one example, the length of the glass substrate **50** in the X-axis direction is approximately 20 mm, the length of the glass substrate **50** in the Y-axis direction is approximately 10 mm, and the thickness of the glass substrate **50** is approximately 0.5 mm.

(33) The glass cover **5** has a first surface **5a** and a second surface **5b**. The first surface **5a** is a surface on the light-incident side of the glass cover **5**. The second surface **5b** is a surface on an opposite side of the glass cover **5** from the first surface **5a**. In the present embodiment, the first surface **5a** is a surface **50a** on the light-incident side of the glass substrate **50**. The second surface **5b** is a surface **50b** on an opposite side of the glass substrate **50** from the first surface **5a**.

(34) The glass cover **5** is fixed above the light-receiving element **3** such that the second surface **5b** faces the light-receiving element **3**. The glass cover **5** is fixed above the light-receiving element **3** in a state where the glass cover **5** is separated from the light-receiving element **3**. More specifically, the second surface **5b** of the glass cover **5** is fixed to the surface **4a** of the frame **4** in a state where the glass cover **5** closes an opening on the light-incident side of the frame **4**, so that the glass cover **5** is fixed above the light-receiving element **3**. The glass cover **5** is fixed to the surface **4a** of the frame **4** by, for example, a resin-based adhesive. As one example, a gap of, for example, approximately 0.5 mm is provided between the light-receiving element **3** and the glass cover **5**.

(35) In the light-receiving device **1A**, the printed wiring substrate **2**, the frame **4**, and the glass cover **5** form a package that houses the light-receiving element **3** and the electronic component **6**. The package defines a space in which the light-receiving element **3** and the electronic component **6** are disposed.

(36) The glass cover **5** includes a plurality of diffraction gratings **51**. The plurality of diffraction gratings **51** are provided in the first surface **5a** of the glass cover **5**. In the present embodiment, the plurality of diffraction gratings **51** are formed in the surface **50a** of the glass substrate **50**. The plurality of diffraction gratings **51** correspond one-to-one to the plurality of light-receiving portions **3a**. The diffraction grating **51** and the light-receiving portion **3a** corresponding to each other at least partially overlap each other when viewed in the Z-axis direction. Each of the diffraction gratings **51** is configured to concentrate the laser light **L1** for distance measurement (in the present embodiment, the laser light **L1** for distance measurement that is emitted from the light source **10** and is reflected by the object **A**) on each of the light-receiving portions **3a**, the laser light **L1** incident on the first surface **5a** of the glass cover **5** in parallel with the Z-axis direction. Namely, each of the diffraction gratings **51** is configured to concentrate light, which has a wavelength corresponding to the wavelength of the laser light **L1** for distance measurement, on each of the light-receiving portions **3a**.

(37) As illustrated in FIG. **3**, the diffraction grating **51** is formed in, for example, a rectangular shape. In the present embodiment, the diffraction grating **51** is formed of a plurality of refractive index change areas **51a**, in which the refractive index is changed from that of a surrounding area, on the surface **50a** of the glass substrate **50**. The transmittance of the laser light **L1** for distance measurement in the refractive index change areas **51a** is lower than the transmittance of the laser light **L1** for distance measurement in an area other than the refractive index change areas **51a** on the surface **50a** of the glass substrate **50**. As one example, each of the refractive index change areas **51a** extends along the Y-axis direction, and the distance between the adjacent refractive index change areas **51a** decreases from the center toward both sides in the X-axis direction. Accordingly, the diffraction grating **51** concentrates the laser light **L1** for distance measurement, which has been incident in parallel with the Z-axis direction, only in a plane perpendicular to the Y-axis direction. Therefore, the laser light **L1** for distance measurement that has been concentrated by the diffraction grating **51** has an irradiation area **R1**, which has the Y-axis direction as a longitudinal direction, on

the corresponding light-receiving portion **3a**. As one example, the width of the irradiation area **R1** in the X-axis direction is smaller than the width of the light-receiving portion **3a** in the X-axis direction, and the width of the irradiation area **R1** in the Y-axis direction is larger than the width of the light-receiving portion **3a** in the Y-axis direction.

(38) As described above, in the light-receiving device **1A**, the glass cover **5** includes the plurality of diffraction gratings **51**, and each of the diffraction gratings **51** is configured to concentrate the laser light **L1** for distance measurement on each of the light-receiving portions **3a**, the laser light **L1** for distance measurement incident on the glass cover **5**. Accordingly, a reduction in crosstalk between the adjacent light-receiving portions **3a** and an improvement in sensitivity of each of the light-receiving portions **3a** can be achieved. Moreover, the plurality of diffraction gratings **51** are provided in the first surface **5a** of the glass cover **5**. Accordingly, it is not necessary to provide, as a separate member, a light-concentrating element between the light-receiving element **3** and the glass cover **5**. In addition, since the distance from the light-receiving element **3** to the plurality of diffraction gratings **51** is larger than that when the plurality of diffraction gratings **51** are provided in the second surface **5b** of the glass cover **5**, the glass cover **5** can be fixed above the light-receiving element **3** in a state where the glass cover **5** is placed closer to the light-receiving element **3**. As a result, the light-receiving device **1A** can accurately receive the laser light **L1** for distance measurement, and can be made thin (made thin in the Z-axis direction).

(39) Particularly, in the light-receiving device **1A**, since it can be considered that the laser light **L1** for distance measurement is incident on the first surface **5a** of the glass cover **5** in parallel with the Z-axis direction, the plurality of diffraction gratings **51** having the same pattern as each other can be adopted as a plurality of light-concentrating elements. Namely, in the light-receiving device **1A**, since light having a wavelength corresponding to the wavelength of the laser light **L1** for distance measurement and incident on the first surface **5a** of the glass cover **5** in parallel with the Z-axis direction can be effectively concentrated on each of the light-receiving portions **3a**, the plurality of diffraction gratings **51** having the same pattern as each other can be easily and accurately formed.

(40) Incidentally, the light-receiving device **1A** may be used under an environment in which a change in temperature is large, such as a case in which the distance measurement system **100** is used for in-vehicle applications. In such a case, it is very effective to provide, as a plurality of light-concentrating elements, the plurality of diffraction gratings **51** in the glass cover **5**. The reason is that the coefficient of thermal expansion of the glass material is closer to the coefficient of thermal expansion of a semiconductor material forming the light-receiving element **3** than a resin material, and the glass material has higher environmental resistance than the resin material. Since the plurality of diffraction gratings **51** are provided, as a plurality of light-concentrating elements, in the glass cover **5**, even under an environment in which a change in temperature is large, the laser light **L1** for distance measurement incident on the glass cover **5** can be reliably concentrated on each of the light-receiving portions **3a**.

(41) In addition, in the light-receiving device **1A**, the glass cover **5** is fixed above the light-receiving element **3** in a state where the glass cover **5** is separated from the light-receiving element **3**. In addition, in the light-receiving device **1A**, the plurality of diffraction gratings **51** are formed in the surface **50a** of the glass substrate **50** forming the glass cover **5**. Accordingly, the distance from the light-receiving element **3** to the plurality of diffraction gratings **51** can be set to an appropriate distance, and the laser light **L1** for distance measurement incident on the glass cover **5** can be reliably concentrated on each of the light-receiving portions **3a**.

(42) Next, a method for manufacturing the light-receiving device **1A** will be described. First, as illustrated in FIG. 4, the light-receiving element **3** is prepared in a state where the light-receiving element **3** and the electronic component **6** are mounted on the printed wiring substrate **2** and the frame **4** is fixed to the printed wiring substrate **2** (first step). Meanwhile, the glass cover **5** is prepared (first step). Subsequently, as illustrated in FIG. 5, the frame **4** is fixed to the glass cover **5**. Therefore, the glass cover **5** is fixed above the light-receiving element **3** such that the glass cover **5**

is separated from the light-receiving element 3 (second step).

(43) Subsequently, as illustrated in FIG. 6, the plurality of diffraction gratings 51 are formed in the glass cover 5 by irradiating the glass cover 5 with laser light L2 for processing (third step). In the present embodiment, the plurality of diffraction gratings 51 are formed in the surface 50a of the glass substrate 50 such that each of the diffraction gratings 51 concentrates the laser light L1 for distance measurement on each of the light-receiving portions 3a, the laser light L1 for distance measurement incident on the glass cover 5. The surface 50a of the glass substrate 50 is irradiated with the laser light L2 for processing from a side opposite to the light-receiving element 3 along the Z-axis direction in a state where a concentration point is placed on the surface 50a of the glass substrate 50. The concentration point of the laser light L2 for processing is scanned on the surface 50a of the glass substrate 50, for example, with reference to the light-receiving portion 3a. The laser light L2 for processing is, for example, femtosecond laser light. The light-receiving device 1A is obtained from the above steps.

(44) As described above, in the method for manufacturing the light-receiving device 1A, the plurality of diffraction gratings 51 are formed in the glass cover 5 by fixing the glass cover 5 above the light-receiving element 3 and then irradiating the glass cover 5 with the laser light L2 for processing. Accordingly, the light-receiving device 1A in which the diffraction gratings 51 are accurately positioned with respect to the respective light-receiving portions 3a can be efficiently obtained. As a result, according to the method for manufacturing the light-receiving device 1A, the light-receiving device 1A can be appropriately manufactured.

(45) In addition, in the method for manufacturing the light-receiving device 1A, the glass cover 5 is fixed above the light-receiving element 3 such that the glass cover 5 is separated from the light-receiving element 3. Accordingly, in the manufactured light-receiving device 1A, the distance from the light-receiving element 3 to the plurality of diffraction gratings 51 can be set to an appropriate distance, and the laser light L1 for distance measurement incident on the glass cover 5 can be reliably concentrated on each of the light-receiving portions 3a.

(46) In addition, in the method for manufacturing the light-receiving device 1A, the plurality of diffraction gratings 51 are formed in the surface 50a of the glass substrate 50. Accordingly, the influence of the irradiation of the laser light L2 for processing on the light-receiving element 3 can be suppressed.

(47) In addition, in the method for manufacturing the light-receiving device 1A, the plurality of diffraction gratings 51 are formed as a plurality of light-concentrating elements. Accordingly, the plurality of light-concentrating elements can be easily provided in the glass cover 5 by the irradiation of the laser light L2 for processing.

Second Embodiment

(48) As illustrated in FIG. 7, a light-receiving device 1B mainly differs from the light-receiving device 1A described above in that the glass cover 5 is fixed on the light-receiving element 3 in a state where the glass cover 5 is in contact with the light-receiving element 3. In the light-receiving device 1B, the glass cover 5 is fixed on the light-receiving element 3 by, for example, a resin having transmission properties or a resin sheet (adhesive layer) having a film structure. In the present embodiment, when viewed in the Z-axis direction, the glass cover 5 is formed in a long shape in which an outer edge of the glass cover 5 is aligned with an outer edge of the light-receiving element 3. Also in the light-receiving device 1B, the plurality of diffraction gratings 51 are provided in the first surface 5a of the glass cover 5, and each of the diffraction gratings 51 is configured to concentrate the laser light L1 for distance measurement on each of the light-receiving portions 3a, the laser light L1 for distance measurement incident on the first surface 5a of the glass cover 5 in parallel with the Z-axis direction. Incidentally, a configuration in which only an adhesive layer exists between the light-receiving element 3 and the glass cover 5 corresponds to a configuration in which the glass cover 5 is fixed on the light-receiving element 3 in a state where the glass cover 5 is in contact with the light-receiving element 3.

(49) The light-receiving device 1B described above can be made thinner (made thin in the Z-axis direction) in addition to exhibiting the same effects as those of the light-receiving device 1A described above. Particularly, in the light-receiving device 1B, since the glass cover 5 is fixed on the light-receiving element 3 in a state where the glass cover 5 is in contact with the light-receiving element 3, the distance from the light-receiving element 3 to the plurality of diffraction gratings 51 can be more accurately defined than, for example, when the glass cover 5 is fixed above the light-receiving element 3 with another member interposed therebetween. Incidentally, in the configuration in which the glass cover 5 is fixed on the light-receiving element 3 in a state where the glass cover 5 is in contact with the light-receiving element 3, it is more important that a cover including the plurality of diffraction gratings 51 forms the glass substrate 50. The reason is that generally, the coefficient of thermal expansion of glass is closer to the coefficient of thermal expansion of the semiconductor material forming the light-receiving element 3 than the coefficient of thermal expansion of the resin, and even in an environment in which a change in temperature is large (for example, an interior or the like of an automobile), misalignment between the diffraction grating 51 and the light-receiving portion 3a corresponding to each other is unlikely to occur.

(50) Next, a method for manufacturing the light-receiving device 1B will be described. First, as illustrated in FIG. 8, a semiconductor wafer 3W including a plurality of portions that become the light-receiving elements 3, and a glass wafer 5W including a plurality of portions (not illustrated) that become the glass covers 5 are prepared (first step). In the semiconductor wafer 3W, the plurality of portions that become a plurality of the light-receiving elements 3 are arranged in two dimensions. The glass wafer 5W is sized to cover the plurality of portions which become the plurality of light-receiving elements 3. Subsequently, as illustrated in FIG. 9, for example, the glass wafer 5W is fixed on the semiconductor wafer 3W by, for example, a resin having transmission properties or a resin sheet having a film structure (second step). This step corresponds to fixing the glass cover 5 on the light-receiving element 3 such that the glass cover 5 is in contact with the light-receiving element 3.

(51) Subsequently, as illustrated in FIG. 10, the plurality of diffraction gratings 51 are formed for each of the portions, which becomes the light-receiving element 3, by irradiating the glass wafer 5W with the laser light L2 for processing (third step). In the present embodiment, the plurality of diffraction gratings 51 are formed in a surface 5Wa of the glass wafer 5W such that each of the diffraction gratings 51 concentrates the laser light L1 for distance measurement on each of the light-receiving portions 3a, the laser light L1 for distance measurement incident on the glass cover 5. This step corresponds to forming the plurality of diffraction gratings 51 in the surface 50a of the glass substrate 50. The surface 5Wa is a surface on an opposite side of the glass wafer 5W from the semiconductor wafer 3W. The surface 5Wa of the glass wafer 5W is irradiated with the laser light L2 for processing from a side opposite to the semiconductor wafer 2W along the Z-axis direction in a state where a concentration point is placed on the surface 5Wa of the glass wafer 5W. The concentration point of the laser light L2 for processing is scanned on the surface 5Wa of the glass wafer 5W, for example, with reference to the light-receiving portion 3a for each of the portions that become the light-receiving elements 3. The laser light L2 for processing is, for example, femtosecond laser light. Subsequently, as illustrated in FIG. 11, the semiconductor wafer 3W and the glass wafer 5W are cut for each of the portions that become the light-receiving elements 3 (fourth step). A plurality of the light-receiving devices 1B are obtained from the above steps.

(52) According to the method for manufacturing the light-receiving device 1B described above, the following effect is exhibited in addition to exhibiting the same effects as those in the method for manufacturing the light-receiving device 1A described above. Namely, according to the method for manufacturing the light-receiving device 1B, the manufactured light-receiving device 1B can be made thinner. In addition, since the method for manufacturing the light-receiving device 1B is performed at the wafer level, the plurality of diffraction gratings 51 can be efficiently formed by the irradiation of the laser light L2 for processing, and as a result, the plurality of light-receiving

devices **1B** can be efficiently manufactured.

Third Embodiment

(53) As illustrated in FIG. **12**, a light-receiving device **1C** of a third embodiment mainly differs from the light-receiving device **1A** described above in that a through-hole **20** is formed in the printed wiring substrate **2**. In the light-receiving device **1C**, the printed wiring substrate (substrate) **2**, the frame **4**, and the glass cover **5** form a package **P** that houses the light-receiving element **3**. Namely, the printed wiring substrate **2** forms, together with the glass cover **5**, the package **P** that houses the light-receiving element **3**.

(54) FIG. **13(a)** is a cross-sectional view of a portion of the printed wiring substrate **2**, which includes the through-hole **20**, taken in parallel with the Y-axis direction, and FIG. **13(b)** is a cross-sectional view of a portion of the printed wiring substrate **2**, which includes the through-hole **20**, taken in parallel with the X-axis direction. As illustrated in FIGS. **13(a)** and **13(b)**, the through-hole **20** is open to the inside surface **2a** and an outside surface **2b** of the printed wiring substrate **2**. The inside surface **2a** is a surface on which the light-receiving element **3** (refer to FIG. **12**) is disposed, and the outside surface **2b** is a surface opposite to the inside surface **2a**. The through-hole **20** is a tapered hole that widens from the inside surface **2a** toward the outside surface **2b**. The through-hole **20** includes an inside opening portion **21**, an outside opening portion **22**, and an intermediate opening portion **23**. The inside opening portion **21** is located in the inside surface **2a**. The outside opening portion **22** is located in the outside surface **2b**. The intermediate opening portion **23** is located between the inside surface **2a** and the outside surface **2b**.

(55) As illustrated in FIG. **14**, the outside opening portion **22** has a shape through which a metallic ball **8** to be described later is passable. Namely, when viewed in the Z-axis direction, the outside opening portion **22** has a shape which can include the entirety of the metallic ball **8** at any position. The intermediate opening portion **23** has a shape through which the metallic ball **8** is not passable. Namely, when viewed in the Z-axis direction, the intermediate opening portion **23** has a shape which cannot include the entirety of the metallic ball **8** at any position. In the present embodiment, the intermediate opening portion **23** has a long shape having one direction as a longitudinal direction (namely, a shape in which the width in the one direction is larger than the width in a direction perpendicular to the one direction). When viewed in the Z-axis direction, both end portions of the intermediate opening portion **23** in the longitudinal direction do not overlap the metallic ball **8**. The shape of the inside opening portion **21** is arbitrary. Incidentally, the Z-axis direction is a direction in which the inside surface **2a** and the outside surface **2b** face each other.

(56) As one example, the inside opening portion **21** has a circular shape, and each of the outside opening portion **22** and the intermediate opening portion **23** has an oval shape having the X-axis direction as a longitudinal direction. When viewed in the Z-axis direction, the inside opening portion **21** is located inside the intermediate opening portion **23**, and the intermediate opening portion **23** is located inside the outside opening portion **22**. Incidentally, in FIG. **14**, a metallic film **7** and a solder member **9** to be described later are not illustrated.

(57) As illustrated in FIGS. **13(a)** and **13(b)**, the metallic film **7** is provided on an inner surface **20a** of the through-hole **20**. The metallic film **7** includes a cylindrical portion **71** and a flange portion **72**. The cylindrical portion **71** is provided on a side surface **20b** of the inner surface **20a** of the through-hole **20**, the side surface **20b** being located between the outside opening portion **22** and the intermediate opening portion **23**. The flange portion **72** is provided on the outside surface **2b** of the printed wiring substrate **2** to extend along an outer edge of the outside opening portion **22**. The cylindrical portion **71** is connected to the flange portion **72**. In the present embodiment, the metallic film **7** reaches the inside opening portion **21**. The metallic film **7** is formed, for example, by Cu or Au plating.

(58) The metallic ball **8** is disposed in the through-hole **20**. The metallic ball **8** is disposed in the through-hole **20** to be surrounded by the cylindrical portion **71** of the metallic film **7**. More specifically, a portion on an outside opening portion **22** side of the metallic ball **8** is surrounded by

the cylindrical portion 71 of the metallic film 7. The metallic ball 8 has a spherical shape. The center of the metallic ball 8 is located between the outside opening portion 22 and the intermediate opening portion 23. In the present embodiment, the metallic ball 8 closes a part of the intermediate opening portion 23 (portion between both the end portions of the intermediate opening portion 23 in the longitudinal direction). The metallic ball 8 is made of, for example, copper.

(59) The solder member 9 is disposed in the through-hole 20. The solder member 9 is disposed between the metallic film 7 and the metallic ball 8 in the through-hole 20, and fixes the metallic ball 8 in the through-hole 20. More specifically, the solder member 9 seals an area between the cylindrical portion 71 of the metallic film 7 and the metallic ball 8 in a state where the solder member 9 extends from the metallic ball 8 to the cylindrical portion 71 of the metallic film 7. When viewed in the Z-axis direction, the solder member 9 extends in an annular shape in the area between the cylindrical portion 71 of the metallic film 7 and the metallic ball 8. A surface 9a on the outside opening portion 22 side of the solder member 9 is a curved surface that is recessed to an inside opening portion 21 side in the area between the cylindrical portion 71 of the metallic film 7 and the metallic ball 8 (namely, has a fillet shape). In the present embodiment, the solder member 9 reaches the inside opening portion 21, and does not reach the outside opening portion 22. The solder member 9 is made of, for example, SnAgCu or AuSn.

(60) A method for manufacturing the light-receiving device 1C configured as described above will be described. Incidentally, FIG. 15(a) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the Y-axis direction, and FIG. 15(b) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the X-axis direction. In addition, FIG. 16(a) is a cross-sectional view of the portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the Y-axis direction, and FIG. 16(b) is a cross-sectional view of the portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the X-axis direction.

(61) First, the light-receiving element 3 and the like are disposed on the inside surface 2a of the printed wiring substrate 2 by performing the first step and the second step in the method for manufacturing the light-receiving device 1A described above, so that the package P is formed by the printed wiring substrate (substrate) 2, the frame 4, and the glass cover 5 (fifth step). Before the light-receiving element 3 and the like are disposed on the inside surface 2a of the printed wiring substrate 2, the through-hole 20 is formed in the printed wiring substrate 2, and the metallic film 7 is provided on the inner surface 20a of the through-hole 20.

(62) Subsequently, as illustrated in FIGS. 15(a) and 15(b), the package P (refer to FIG. 12) is disposed such that the outside surface 2b is located on an upper side with respect to the inside surface 2a, and a solder ball S is disposed in the through-hole 20 from the outside opening portion 22 (sixth step). The solder ball S is a solder ball with a core, which includes the metallic ball 8 and a solder layer 90 covering the metallic ball 8. The melting point of the metallic ball 8 is higher than the melting point of the solder layer 90. Subsequently, as illustrated in FIGS. 16(a) and 16(b), the solder member 9 is formed by melting the solder layer 90 (sixth step). In the present embodiment, as illustrated in FIGS. 15(a) and 15(b), the solder layer 90 is melted by irradiating the solder ball S with laser light L while degassing the inside of the package P through the through-hole 20.

(63) Subsequently, the light-receiving device 1C is obtained by performing the third step in the method for manufacturing the light-receiving device 1A described above. Incidentally, after the third step in the method for manufacturing the light-receiving device 1A described above is performed, the sixth step described above may be performed.

(64) A method for mounting the light-receiving device 1C manufactured as described above will be described. Incidentally, FIG. 17(a) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the Y-axis direction, and FIG. 17(b) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes

the through-hole **20**, taken in parallel with the X-axis direction. In addition, FIG. **18(a)** is a cross-sectional view of the portion of the printed wiring substrate **2**, which includes the through-hole **20**, taken in parallel with the Y-axis direction, and FIG. **18(b)** is a cross-sectional view of the portion of the printed wiring substrate **2**, which includes the through-hole **20**, taken in parallel with the X-axis direction.

(65) First, as illustrated in FIGS. **17(a)** and **17(b)**, a mounting substrate **200** on which the light-receiving device **1C** is to be mounted is prepared. The mounting substrate **200** is provided with a metallic pattern **201** corresponding to the outside opening portion **22** of the through-hole **20** and the flange portion **72** of the metallic film **7** in the light-receiving device **1C**, and a solder layer **202** is provided on the metallic pattern **201**. Subsequently, the printed wiring substrate **2** is disposed on the mounting substrate **200** such that the outside opening portion **22** and the flange portion **72** are located on the metallic pattern **201** with the solder layer **202** interposed therebetween.

(66) Subsequently, as illustrated in FIGS. **18(a)** and **18(b)**, the flange portion **72** of the metallic film **7** is joined to the metallic pattern **201**, for example, by melting the solder layer **202** in a heating furnace. At this time, when the solder member **9** (refer to FIGS. **17(a)** and **17(b)**) is melted, the molten solder member **9** contributes to the joining between the flange portion **72** and the metallic pattern **201**. In addition, since the solder member **9** is melted, even when the metallic ball **8** falls on the metallic pattern **201**, the metallic ball **8** is fixed on the metallic pattern **201** by the solder layer **202** and the solder member **9**.

(67) As described above, according to the light-receiving device **1C**, during manufacturing of the light-receiving device **1C**, for example, while degassing the inside of the package **P** through the through-hole **20**, the metallic ball **8** can be fixed in the through-hole **20** by the solder member **9**, and the area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8** can be sealed by the solder member **9**. At this time, since at least a part of the intermediate opening portion **23** of the through-hole **20** is closed by the metallic ball **8**, the molten solder member **9** can be suppressed from infiltrating into the package **P**. Further, the intermediate opening portion **23** of the through-hole **20** has a shape through which the metallic ball **8** is not passable, and in a state where the solder member **9** extends from the metallic ball **8** to the cylindrical portion **71** of the metallic film **7**, the area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8** is sealed by the solder member **9**. Accordingly, in the manufactured light-receiving device **1C**, the fixed state of the metallic ball **8** is stabilized, so that the sealing state of the through-hole **20** can be reliably maintained.

(68) In addition, in the light-receiving device **1C**, the intermediate opening portion **23** of the through-hole **20** has a long shape. Accordingly, during manufacturing of the light-receiving device **1C**, when the metallic ball **8** is fixed in the through-hole **20** by the solder member **9**, a part of the intermediate opening portion **23** of the through-hole **20** (in the present embodiment, both the end portions of the intermediate opening portion **23** in the longitudinal direction) is not closed by the metallic ball **8**, so that the inside of the package **P** can be reliably degassed through the through-hole **20**.

(69) In addition, in the light-receiving device **1C**, the through-hole **20** is a tapered hole that widens from the inside surface **2a** toward the outside surface **2b**. Accordingly, the through-hole **20** including the outside opening portion **22** having a shape through which the metallic ball **8** is passable and the intermediate opening portion **23** having a shape through which the metallic ball **8** is not passable can be easily and reliably obtained.

(70) In addition, in the light-receiving device **1C**, the metallic film **7** includes the cylindrical portion **71** and the flange portion **72**, and the cylindrical portion **71** is connected to the flange portion **72**. Accordingly, for example, when the flange portion **72** of the metallic film **7** is joined to the metallic pattern **201** of the mounting substrate **200**, even if the metallic ball **8** falls on the metallic pattern **201**, the sealing state of the through-hole **20** in the light-receiving device **1C** after mounting can be reliably maintained.

(71) In addition, in the light-receiving device **1C**, the surface **9a** of the solder member **9** is a curved surface that is recessed to the inside opening portion **21** side in the area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8**. Accordingly, the local concentration of stress can be suppressed from occurring in the solder member **9**, and the sealing state of the through-hole **20** can be more reliably maintained.

(72) In addition, according to the method for manufacturing the light-receiving device **1C**, the through-hole **20** can be reliably sealed while suppressing the infiltration of the molten solder layer **90** into the package **P**.

(73) In addition, in the method for manufacturing the light-receiving device **1C**, the solder layer **90** is melted by irradiating the solder ball **S** with the laser light **L**. Accordingly, the solder layer **90** can be locally heated, so that thermal influence on the light-receiving device **1C** and the like can be suppressed.

(74) In addition, in the method for manufacturing the light-receiving device **1C**, the solder layer **90** is melted while the inside of the package **P** is degassed through the through-hole **20**. Accordingly, the inside of the package **P** can be reliably degassed through the through-hole **20** while suppressing the infiltration of the molten solder layer **90** into the package.

Modification Examples

(75) The present disclosure is not limited to the first embodiment, the second embodiment, and the third embodiment. For example, the glass substrate **50** is not limited to being disposed such that the surface **50a** and the surface **50b** are parallel with each other, and may be disposed such that the surface **50a** is inclined with respect to the surface **50b**. In each of the light-receiving devices **1A**, **1B**, and **1C**, the plurality of diffraction gratings **51** may be formed inside the glass substrate **50**. Also in this case, the influence of the irradiation of the laser light **L2** for processing on the light-receiving element **3** can be suppressed. In addition, in each of the light-receiving devices **1A**, **1B**, and **1C**, as illustrated in FIG. **19**, the glass cover **5** may include the glass substrate **50** and a first film **52** provided on the surface **50a** of the glass substrate **50**, and the plurality of diffraction gratings **51** may be formed in the first film **52**. Accordingly, the distance from the light-receiving element **3** to the plurality of diffraction gratings **51** can be set to an appropriate distance, and the laser light **L1** for distance measurement incident on the glass cover **5** can be reliably concentrated on each of the plurality of light-receiving portions **3a**. In the example illustrated in FIG. **19**, a plurality of slits formed in the first film **52** form each of the diffraction gratings **51**. In this case, in each of the diffraction gratings **51**, an area in each slit corresponds to a refractive index change area, in which the refractive index is changed from a surrounding area, in the first film **52**. Namely, in the example illustrated in FIG. **19**, the refractive index change areas in which the transmittance of the laser light **L1** for distance measurement is higher than in the first film **52** form the diffraction grating **51**. The first film **52** is, for example, a SiN film.

(76) In addition, in each of the light-receiving devices **1A**, **1B**, and **1C**, the light-receiving element **3** may be an area sensor including the plurality of light-receiving portions **3a** arranged in two dimensions.

(77) As one example, when the plurality of light-receiving portions **3a** are arranged in a matrix pattern having the X-axis direction as a row direction and the Y-axis direction as a column direction, each of the diffraction gratings **51** may include a first layer including the plurality of refractive index change areas **51a** extending along the X-axis direction and a second layer including the plurality of refractive index change areas **51a** extending along the Y-axis direction. In the first layer, when the plurality of refractive index change areas **51a** are formed such that the distance between the adjacent refractive index change areas **51a** decreases from the center toward both sides in the Y-axis direction, the laser light **L1** for distance measurement incident in parallel with the Z-axis direction can be concentrated in a plane perpendicular to the X-axis direction. In the second layer overlapping the first layer when viewed in the Z-axis direction, when the plurality of refractive index change areas **51a** are formed such that the distance between the adjacent

refractive index change areas **51a** decreases from the center toward both sides in the X-axis direction, the laser light **L1** for distance measurement incident in parallel with the Z-axis direction can be concentrated in a plane perpendicular to the Y-axis direction. Therefore, the laser light **L1** for distance measurement can be reliably concentrated on each of the plurality of light-receiving portions **3a** arranged in a matrix pattern.

(78) In addition, as one example, when the plurality of light-receiving portions **3a** are arranged in a matrix pattern having the X-axis direction as a row direction and the Y-axis direction as a column direction, each of the diffraction gratings **51** may be formed of the plurality of refractive index change areas **51a** that are concentrically formed. When the plurality of refractive index change areas **51a** are formed such that the distance between the adjacent refractive index change areas **51a** decreases from the center toward an outer side, the laser light **L1** for distance measurement can be reliably concentrated on each of the plurality of light-receiving portions **3a** arranged in a matrix pattern.

(79) In addition, in the light-receiving device **1B**, the glass cover **5** may be fixed on the light-receiving element **3** by direct bonding. In this case, in the method for manufacturing the light-receiving device **1B**, in the second step, the glass wafer **5W** is fixed on the semiconductor wafer **3W** by direct bonding. Examples of the direct bonding include plasma activation bonding, surface-activated room-temperature bonding (SAB), atomic diffusion bonding (ADB), anodic bonding, fusion bonding, hydrophilic bonding, and the like.

(80) In addition, each of the light-receiving devices **1A**, **1B**, and **1C** may include the light source **10** that emits the laser light **L1** for distance measurement. Accordingly, the distance to the object **A** can be accurately measured.

(81) In addition, in the method for manufacturing each of the light-receiving devices **1A**, **1B**, and **1C**, as illustrated in FIG. **19**, when the glass cover **5** includes the glass substrate **50** and the first film **52** provided on the surface **50a** of the glass substrate **50**, the plurality of diffraction gratings **51** may be formed in the first film **52** by irradiating the glass cover **5** with the laser light **L2** for processing. Accordingly, while suppressing the influence of the irradiation of the laser light **L2** for processing on the light-receiving devices **1A**, **1B**, and **1C**, a plurality of light-concentrating elements can be easily provided in the glass cover **5** by the irradiation of the laser light **L2** for processing.

(82) In addition, in each of the light-receiving devices **1A**, **1B**, and **1C**, as illustrated in FIG. **20**, the glass cover **5** may further include a filter **54** disposed on a side opposite to the light-receiving element **3** with respect to the plurality of diffraction gratings **51**. The filter **54** may have transmission properties for the laser light **L1** for distance measurement, and have light shielding properties for light having wavelengths other than the wavelength of the laser light **L1** for distance measurement. In the filter **54**, the transmittance of the laser light **L1** for distance measurement is higher than the transmittance of the light having the wavelengths other than the wavelength of the laser light **L1** for distance measurement. In such a manner, since the filter **54** is provided in a previous stage of the plurality of diffraction gratings **51**, the light having the wavelengths other than the wavelength of the laser light **L1** for distance measurement can be reliably cut, and the SN ratio of the light-receiving element **3** can be improved. As one example, the filter **54** is a bandpass filter having transmission properties for the laser light **L1** for distance measurement and having light shielding properties for light having wavelengths other than the wavelength of the laser light **L1** for distance measurement.

(83) In addition, in the method for manufacturing each of the light-receiving devices **1A**, **1B**, and **1C**, the plurality of diffraction gratings (light-concentrating elements) **51** may be formed inside the glass substrate **50** by irradiating the glass cover **5** with the laser light **L2** for processing. In addition, instead of the plurality of diffraction gratings **51**, as illustrated in FIG. **21**, a plurality of lens surfaces (light-concentrating elements) **53** may be formed. In the example illustrated in FIG. **21**, each of the lens surfaces **53** is a cylindrical lens surface that concentrates the laser light **L1** for

distance measurement, which is incident in parallel with the Z-axis direction, only in a plane perpendicular to the Y-axis direction. In addition, instead of the plurality of diffraction gratings 51, as illustrated in FIG. 22, a plurality of light-guiding portions (light-concentrating elements) 55 may be formed. In the example illustrated in FIG. 22, each of the light-guiding portions 55 is formed of a pair of light-guiding areas 55a which are formed to face each other in the X-axis direction and such that the interval therebetween decreases as the pair of light-guiding portions 55 approach the light-receiving portion 3a. In such a manner, in the method for manufacturing each of the light-receiving devices 1A, 1B, and 1C, appropriate light-concentrating elements according to the application may be provided in the glass cover 5. Incidentally, each of the above-described light-concentrating elements is formed of a refractive index change area, in which the refractive index is changed from a surrounding area, inside the glass substrate 50.

(84) In addition, in the method for manufacturing each of the light-receiving devices 1A, 1B, and 1C, as illustrated in FIGS. 23(a) and 23(b), the glass cover 5 may include the glass substrate 50 and a second film 56 provided on the surface 50b of the glass substrate 50. The second film 56 may have transmission properties for the laser light L1 for distance measurement and have light shielding properties for the laser light L2 for processing. In the second film 56, the transmittance of the laser light L1 for distance measurement is higher than the transmittance of the laser light L2 for processing. Accordingly, the influence of the irradiation of the laser light L2 for processing on the light-receiving element 3 can be more reliably suppressed.

(85) Incidentally, the second film 56 may be a film that absorbs the laser light L2 for processing, or a film that reflects the laser light L2 for processing. As one example, the second film 56 is a bandpass filter having transmission properties for the laser light L1 for distance measurement and having light shielding properties for the laser light L2 for processing.

(86) In addition, in the method for manufacturing each of the light-receiving devices 1A, 1B, and 1C, the glass cover 5 may include the glass substrate 50, and the glass substrate 50 may have transmission properties for the laser light L1 for distance measurement and have light shielding properties for the laser light L2 for processing. In this case, the transmittance of the laser light L1 for distance measurement in the glass substrate 50 is higher than the transmittance of the laser light L2 for processing in the glass substrate 50. Accordingly, the influence of the irradiation of the laser light L2 for processing on the light-receiving element 3 can be more reliably suppressed. As one example, the glass substrate 50 is a glass substrate containing a pigment having higher absorption for the laser light L2 for processing than for the laser light L1 for distance measurement.

(87) In addition, in the method for manufacturing each of the light-receiving devices 1A, 1B, and 1C, a plurality of light-concentrating elements may be formed in the glass cover 5 by irradiating the glass cover 5 with the laser light L2 for processing in a state where the glass cover 5 is disposed above the light-receiving element 3, and thereafter, the glass cover 5 may be fixed on the light-receiving element 3.

(88) In addition, in the light-receiving device 1C, as illustrated in FIGS. 24(a) and 24(b), the through-hole 20 formed in the printed wiring substrate 2 may be a stepped hole that is widened in width on an outside surface 2b side. Incidentally, FIG. 24(a) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the Y-axis direction, and FIG. 24(b) is a cross-sectional view of a portion of the printed wiring substrate 2, which includes the through-hole 20, taken in parallel with the X-axis direction.

(89) In the example illustrated in FIGS. 24(a) and 24(b) (hereinafter, referred to as the “light-receiving device 1C of a modification example”), a recessed portion 24 formed in the outside surface 2b and a hole portion 25 formed in a bottom surface 24a of the recessed portion 24 form the through-hole 20. In the light-receiving device 1C of the modification example, an opening portion of the hole portion 25 which is located in the inside surface 2a corresponds to the inside opening portion 21 of the through-hole 20, an opening portion of the recessed portion 24 which is located in the outside surface 2b corresponds to the outside opening portion 22 of the through-hole

20, and an opening portion of the hole portion **25** which is located in the bottom surface **24a** of the recessed portion **24** corresponds to the intermediate opening portion **23** of the through-hole **20**.

(90) In such a manner, even when the through-hole **20** is a stepped hole, the through-hole **20** including the outside opening portion **22** having a shape through which the metallic ball **8** is passable and the intermediate opening portion **23** having a shape through which the metallic ball **8** is not passable can be easily and reliably obtained.

(91) The light-receiving device **1C** of the modification example will be described in more detail. As illustrated in FIG. **25**, the outside opening portion **22** has a shape through which the metallic ball **8** is passable. The intermediate opening portion **23** has a shape through which the metallic ball **8** is not passable. In the light-receiving device **1C** of the modification example, the intermediate opening portion **23** has a long shape having one direction as a longitudinal direction. When viewed in the Z-axis direction, both end portions of the intermediate opening portion **23** in the longitudinal direction do not overlap the metallic ball **8**. The shape of the inside opening portion **21** is arbitrary.

(92) As one example, each of the inside opening portion **21** and the intermediate opening portion **23** has an oval shape having the X-axis direction as a longitudinal direction, and the outside opening portion **22** has a circular shape. In the light-receiving device **1C** of the modification example, the inside opening portion **21** and the intermediate opening portion **23** have the same shape. When viewed in the Z-axis direction, the inside opening portion **21** and the intermediate opening portion **23** are located inside the outside opening portion **22**. Incidentally, in FIG. **25**, the metallic film **7** and the solder member **9** are not illustrated.

(93) As illustrated in FIGS. **24(a)** and **24(b)**, the metallic film **7** is provided on the inner surface **20a** of the through-hole **20**. The metallic film **7** includes the cylindrical portion **71** and the flange portion **72**. The cylindrical portion **71** is provided on the side surface **20b** (namely, a side surface **24b** on a recessed portion **24** side) of the inner surface **20a** of the through-hole **20**, the side surface **20b** being located between the outside opening portion **22** and the intermediate opening portion **23**. The flange portion **72** is provided on the outside surface **2b** of the printed wiring substrate **2** to extend along an outer edge of the outside opening portion **22**. The cylindrical portion **71** is connected to the flange portion **72**. In the light-receiving device **1C** of the modification example, the metallic film **7** reaches the inside opening portion **21**.

(94) The metallic ball **8** is disposed in the through-hole **20**. The metallic ball **8** is disposed in the through-hole **20** to be surrounded by the cylindrical portion **71** of the metallic film **7**. More specifically, a portion on an outside opening portion **22** side of the metallic ball **8** is surrounded by the cylindrical portion **71** of the metallic film **7**. The metallic ball **8** has a spherical shape. The center of the metallic ball **8** is located between the outside opening portion **22** and the intermediate opening portion **23**. In the light-receiving device **1C** of the modification example, the metallic ball **8** closes a part of the intermediate opening portion **23** (portion between both the end portions of the intermediate opening portion **23** in the longitudinal direction).

(95) The solder member **9** is disposed in the through-hole **20**. The solder member **9** is disposed between the metallic film **7** and the metallic ball **8** in the through-hole **20**, and fixes the metallic ball **8** in the through-hole **20**. More specifically, the solder member **9** seals an area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8** in a state where the solder member **9** extends from the metallic ball **8** to the cylindrical portion **71** of the metallic film **7**. When viewed in the Z-axis direction, the solder member **9** extends in an annular shape in the area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8**. The surface **9a** on the outside opening portion **22** side of the solder member **9** is a curved surface that is recessed to the inside opening portion **21** side in the area between the cylindrical portion **71** of the metallic film **7** and the metallic ball **8**. In the light-receiving device **1C** of the modification example, the solder member **9** does not reach each of the inside opening portion **21** and the outside opening portion **22**. Meanwhile, a part of the solder member **9** is also disposed in the hole portion **25**.

(96) The same effects as those of the light-receiving device **1C** of the third embodiment described

above are exhibited by the light-receiving device **1C** of the modification example.

(97) Incidentally, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the substrate on which the light-receiving element **3** is disposed may be a substrate other than the printed wiring substrate **2**, such as a substrate made of, for example, ceramic, plastic, or the like. In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the substrate on which the light-receiving element **3** is disposed may include a portion corresponding to the frame **4**.

(98) In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, when the intermediate opening portion **23** has a long shape, the shape of the intermediate opening portion **23** is not limited to an oval shape, and may be an elliptical shape, a rectangular shape, or the like. In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the intermediate opening portion **23** may not have a long shape as long as the intermediate opening portion **23** has a shape through which the metallic ball **8** is not passable. In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the through-hole **20** may be a hole other than a tapered hole and a stepped hole.

(99) In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the metallic film **7** may not include the flange portion **72**. In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the metallic film **7** may not reach the inside opening portion **21**.

(100) In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the solder member **9** may not cover a portion on the outside opening portion **22** side of the metallic ball **8**, or may cover the portion. In addition, in the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, a part of the solder member **9** may exist between the metallic ball **8** and the intermediate opening portion **23**, or the metallic ball **8** may be in contact with the intermediate opening portion **23**.

(101) In addition, in the method for manufacturing each of the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, when the solder layer **90** is melted, the inside of the package **P** may not be degassed through the through-hole **20**. In that case, when the package **P** is formed, damage to the package **P** caused by the expansion of gas inside the package **P** can be suppressed. In addition, in the method for manufacturing each of the light-receiving device **1C** of the third embodiment and the light-receiving device **1C** of the modification example, the solder layer **90** may be melted in a heating furnace.

(102) As a reference, the configurations related to the through-hole **20** are applicable to a substrate in which at least one of a light-receiving element and a light-emitting element is disposed in a package that houses at least one of the light-receiving element and light-emitting element. In that case, an optical device includes at least one of a light-receiving element and a light-emitting element, and a substrate including an inside surface on which at least one of the light-receiving element and the light-emitting element is disposed, an outside surface opposite to the inside surface, and a through-hole being open to the inside surface and the outside surface. The optical device includes a package that houses at least one of the light-receiving element and the light-emitting element, a metallic film provided on an inner surface of the through-hole, a metallic ball disposed in the through-hole, and a solder member that is disposed between the metallic film and the metallic ball in the through-hole and that fixes the metallic ball in the through-hole. In the optical device, the through-hole includes an inside opening portion located in the inside surface, an outside opening portion located in the outside surface, and an intermediate opening portion located between the inside surface and the outside surface. The outside opening portion has a shape through which the metallic ball is passable, and the intermediate opening portion has a shape through which the metallic ball is not passable. The metallic film includes a cylindrical portion

provided on a side surface of the inner surface of the through-hole, the side surface being located between the outside opening portion and the intermediate opening portion. The metallic ball is disposed in the through-hole to be surrounded by the cylindrical portion. A solder member seals an area between the cylindrical portion and the metallic ball in a state where the solder member extends from the metallic ball to the cylindrical portion.

REFERENCE SIGNS LIST

(103) **1A, 1B, 1C**: light-receiving device, **2**: printed wiring substrate (substrate), **2a**: inside surface, **2b**: outside surface, **3**: light-receiving element, **3a**: light-receiving portion, **3W**: semiconductor wafer, **5**: glass cover, **5a**: first surface, **5b**: second surface, **5W**: glass wafer, **7**: metallic film, **8**: metallic ball, **9**: solder member, **9a**: surface, **10**: light source, **20**: through-hole, **20a**: inner surface, **20b**: side surface, **21**: inside opening portion, **22**: outside opening portion, **23**: intermediate opening portion, **50**: glass substrate, **50a**: surface, **50b**: surface, **51**: diffraction grating (light-concentrating element), **52**: first film, **53**: lens surface (light-concentrating element), **54**: filter, **55**: light-guiding portion (light-concentrating element), **56**: second film, **71**: cylindrical portion, **72**: flange portion, **90**: solder layer, **A**: object, **L1**: laser light for distance measurement, **L2**: laser light for processing, **L**: laser light, **P**: package, **S**: solder ball.

Claims

1. A method for manufacturing a light-receiving device, the method comprising: a first step of preparing a light-receiving element in which a plurality of light-receiving portions are provided, and a glass cover having a first surface and a second surface opposite to the first surface; a second step of disposing the glass cover above the light-receiving element such that the second surface faces the light-receiving element, after the first step; and a third step of forming a plurality of light-concentrating elements in the glass cover by irradiating the glass cover with laser light for processing after the second step, wherein in the third step, the plurality of light-concentrating elements are formed such that each of the plurality of light-concentrating elements concentrates laser light for distance measurement on each of the plurality of light-receiving portions, the laser light for distance measurement incident on the first surface of the glass cover.
2. The method for manufacturing a light-receiving device according to claim 1, wherein in the second step, the glass cover is fixed above the light-receiving element such that the glass cover is separated from the light-receiving element.
3. The method for manufacturing a light-receiving device according to claim 1, wherein in the second step, the glass cover is fixed on the light-receiving element such that the glass cover is in contact with the light-receiving element.
4. The method for manufacturing a light-receiving device according to claim 1, wherein the glass cover includes a glass substrate and a first film provided on a surface on a first surface side of the glass substrate, and in the third step, the plurality of light-concentrating elements are formed in the first film.
5. The method for manufacturing a light-receiving device according to claim 4, wherein the plurality of light-concentrating elements are a plurality of diffraction gratings.
6. The method for manufacturing a light-receiving device according to claim 1, wherein the glass cover includes a glass substrate, and in the third step, the plurality of light-concentrating elements are formed in the glass substrate.
7. The method for manufacturing a light-receiving device according to claim 6, wherein in the third step, the plurality of light-concentrating elements are formed in a surface on a first surface side of the glass substrate.
8. The method for manufacturing a light-receiving device according to claim 7, wherein the plurality of light-concentrating elements are a plurality of diffraction gratings.
9. The method for manufacturing a light-receiving device according to claim 6, wherein in the third

step, the plurality of light-concentrating elements are formed inside the glass substrate.

10. The method for manufacturing a light-receiving device according to claim 9, wherein the plurality of light-concentrating elements are a plurality of diffraction gratings, a plurality of lens surfaces, or a plurality of light-guiding portions.

11. The method for manufacturing a light-receiving device according to claim 1, wherein the glass cover includes a glass substrate and a second film provided on a surface on a second surface side of the glass substrate, and the second film has transmission properties for the laser light for distance measurement, and has light shielding properties for the laser light for processing.

12. The method for manufacturing a light-receiving device according to claim 1, wherein the glass cover includes a glass substrate, and the glass substrate has transmission properties for the laser light for distance measurement, and has light shielding properties for the laser light for processing.

13. The method for manufacturing a light-receiving device according to claim 1, wherein in the first step, a semiconductor wafer including a plurality of portions that become the light-receiving elements, and a glass wafer including a plurality of portions that become the glass covers are prepared, in the second step, the glass wafer is fixed on the semiconductor wafer, and in the third step, the plurality of light-concentrating elements are formed for each of the portions, which become the light-receiving elements, by irradiating the glass wafer with the laser light for processing.

14. The method for manufacturing a light-receiving device according to claim 13, further comprising: a fourth step of cutting the semiconductor wafer and the glass wafer for each of the portions, which become the light-receiving elements, after the third step.
